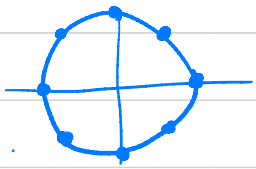
18.05.2021



```
clear; clc; clf

% dhmiourgia kombwn parembolhs
xinterp = [0:pi/4: 2*pi]';
yinterp = sin(xinterp);

%dhmioyrgia shmeiwn gia grafikes parastaseis
xplot = linspace(0, 2*pi, 500);
yplot = sin(xplot); % tetagmenes shmeiwn gia grafikh parastash
plot(xplot, yplot, 'k','LineWidth',2); hold on
plot(xinterp, yinterp, 'ok')

pause
% dhmioyrgia polyonymoy parembolhs
M = vander(xinterp);
p = M\yinterp;

y1=polyval(p, xplot);
plot(xplot, y1,'c-.','LineWidth',2)

pause
% parembolh me grammikh spline
yls=interp1(xinterp,yinterp,xplot,'linear');
plot(xplot, yls,'r-','LineWidth',2)

pause
% parembolh me kybikh spline
ycs=interp1(xinterp,yinterp,xplot,'spline');
plot(xplot, ycs,'g-','LineWidth',2)
```

πολλά σημεία για ομαλές γραφικές παραστάσεις

τα ίδια χρώματα

δημιουργία πίνακα Vandermonde

λύνω σύστημα

$p(x) = c_n x^n + c_{n-1} x^{n-1} + \dots + c_1 x + c_0$

$$M = \begin{bmatrix} x_0^n & x_0^{n-1} & \dots & x_0 & 1 \\ x_1^n & x_1^{n-1} & \dots & x_1 & 1 \\ \vdots & & & & \\ x_n^n & x_n^{n-1} & \dots & x_n & 1 \end{bmatrix}$$

συντελεστές του p.

$$M \cdot C = Y = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix} \Leftrightarrow C = M \backslash Y$$

$$C = \begin{bmatrix} c_n \\ c_{n-1} \\ \vdots \\ c_0 \end{bmatrix}$$